2.16 Trigonometry: Compound Angle Formulae

Worked Solutions

1 a)

$$\cos x = \frac{h}{b} \Rightarrow h = b\cos x$$

1 b)

$$\cos y = \frac{h}{a} \Rightarrow h = a\cos y$$

1 c)

$$\text{Area of triangle } ACM = \frac{1}{2}bh\sin x$$
$$\text{Area of triangle } BCM = \frac{1}{2}ah\sin y$$

1 d)

$$\text{Area of triangle } ABC = \frac{1}{2}ab\sin(x+y)$$

1 e)

$$\text{Area of triangle } ABC = \text{Area of triangle } ACM + \text{Area of triangle } BCM$$
$$\Rightarrow \frac{1}{2}ab\sin(x+y) = \frac{1}{2}bh\sin x + \frac{1}{2}ah\sin y$$
$$\Rightarrow \frac{1}{2}ab\sin(x+y) = \frac{1}{2}b(a\cos y)\sin x + \frac{1}{2}a(b\cos x)\sin y$$
$$\Rightarrow \frac{1}{2}ab\sin(x+y) = \frac{1}{2}ab\sin x\cos y + \frac{1}{2}ab\cos x\sin y$$
$$\Rightarrow \sin(x+y) = \sin x\cos y + \cos x\sin y$$

2 a)

$$AE = \cos x$$
$$CE = \sin x$$

2 b)

$$\angle ACE = 90^\circ - x$$
$$\angle AEF = 90^\circ - y$$
$$\angle CED = y$$
$$\angle ECD = 90^\circ - y$$
$$\angle BCA = x + y$$
$$\angle BAC = 90^\circ - (x+y)$$

2 c)

$$BC = \cos(x+y)$$
$$AB = \sin(x+y)$$

2 d)

$$\cos y = \frac{AF}{AE} \Rightarrow \cos y = \frac{AF}{\cos x} \Rightarrow AF = \cos x \cos y$$
$$\sin y = \frac{EF}{AE} \Rightarrow \sin y = \frac{EF}{\cos x} \Rightarrow EF = \cos x \sin y$$

2 e)

$$\cos y = \frac{DE}{CE} \Rightarrow \cos y = \frac{DE}{\sin x} \Rightarrow DE = \sin x \cos y$$
$$\sin y = \frac{CD}{CE} \Rightarrow \sin y = \frac{CD}{\sin x} \Rightarrow CD = \sin x \sin y$$

2 f)

$$\begin{aligned} AB &= DE + EF \\ &\Rightarrow \sin(x+y) = \sin x \cos y + \cos x \sin y \\ BC &= AF - CD \\ &\Rightarrow \cos(x+y) = \cos x \cos y - \sin x \sin y \end{aligned}$$

3)

$$\begin{aligned} \sin(x-y) &= \sin(x+(-y)) \\ &\equiv \sin x \cos(-y) + \cos x \sin(-y) \end{aligned}$$

We know: $\cos(-y) \equiv \cos y$ and $\sin(-y) \equiv -\sin y$.

$$\begin{aligned} \therefore \sin x \cos(-y) + \cos x \sin(-y) &\equiv \sin x \cos y + \cos x(-\sin y) \\ &\equiv \sin x \cos y - \cos x \sin y \end{aligned}$$

4)

$$\begin{aligned} \cos(x-y) &= \cos(x+(-y)) \\ &\equiv \cos x \cos(-y) - \sin x \sin(-y) \end{aligned}$$

We know: $\cos(-y) \equiv \cos y$ and $\sin(-y) \equiv -\sin y$.

$$\begin{aligned} \therefore \cos x \cos(-y) - \sin x \sin(-y) &\equiv \cos x \cos y - \sin x(-\sin y) \\ &\equiv \cos x \cos y + \sin x \sin y \end{aligned}$$

5)

$$\tan(x+y) \equiv \frac{\sin(x+y)}{\cos(x+y)} \equiv \frac{\sin x\cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y}$$
$$\equiv \frac{\dfrac{\sin x \cos y}{\cos x \cos y} + \dfrac{\cos x \sin y}{\cos x \cos y}}{\dfrac{\cos x \cos y}{\cos x \cos y} - \dfrac{\sin x \sin y}{\cos x \cos y}} \equiv \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

6)

$$\tan(x-y) \equiv \tan(x+(-y)) \equiv \frac{\tan x + \tan(-y)}{1 - \tan x \tan(-y)}$$

We know: $\tan(-y) \equiv \tan y$.

$$\therefore \frac{\tan x + \tan(-y)}{1 - \tan x \tan(-y)} \equiv \frac{\tan x - \tan y}{1 - \tan x(-\tan y)} \equiv \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

7 a)

$$\sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$$
$$\equiv \sin(30^\circ + 45^\circ) \equiv \sin 75^\circ$$

7 b)

$$\sin 35^\circ \cos 20^\circ - \cos 35^\circ \sin 20^\circ$$
$$\equiv \sin(35^\circ - 20^\circ) \equiv \sin 15^\circ$$

7 c)

$$\sin 70^\circ \cos 50^\circ - \cos 70^\circ \sin 50^\circ$$
$$\equiv \sin(70^\circ - 50^\circ) \equiv \sin 20^\circ$$

7 d)

$$\sin 70^\circ \cos 50^\circ + \cos 70^\circ \sin 50^\circ$$
$$\equiv \sin(70^\circ + 50^\circ) \equiv \sin 120^\circ \equiv \sin 60^\circ$$

8 a)

$$\cos 50^\circ \cos 30^\circ + \sin 50^\circ \sin 30^\circ$$
$$\equiv \cos(50^\circ - 30^\circ) \equiv \cos 20^\circ$$

8 b)

$$\cos 68^\circ \cos 13^\circ + \sin 68^\circ \sin 13^\circ$$
$$\equiv \cos(68^\circ - 13^\circ) \equiv \cos 55^\circ$$

8 c)

$$\cos 10^\circ \cos 54^\circ - \sin 10^\circ \sin 54^\circ$$
$$\equiv \cos(10^\circ + 54^\circ) \equiv \cos 64^\circ$$

8 d)

$$\cos 300^\circ \cos 50^\circ - \sin 300^\circ \sin 50^\circ$$
$$\equiv \cos(300^\circ + 50^\circ) \equiv \cos 350^\circ \equiv \cos 10^\circ$$

9 a i)

$$\sin 30^\circ = \frac{1}{2}$$
$$\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$
$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

9 a ii)

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$
$$\cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$
$$\cos 60^\circ = \frac{1}{2}$$

9 b i)

$$\begin{aligned}\sin 75^\circ &\equiv \sin(30^\circ + 45^\circ) \\ &\equiv \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ \\ &\equiv \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} \equiv \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

9 b ii)

$$\begin{aligned}\sin 15^\circ &\equiv \sin(60^\circ - 45^\circ) \\ &\equiv \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ \\ &\equiv \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{1}{2} \times \frac{\sqrt{2}}{2} \equiv \frac{\sqrt{6} - \sqrt{2}}{4}\end{aligned}$$

Alternatively:

$$\begin{aligned}\sin 15^\circ &\equiv \sin(45^\circ - 30^\circ)\\ &\equiv \sin 45^\circ \cos 30^\circ \equiv \cos 45^\circ \sin 30^\circ\\ &\equiv \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \times \frac{1}{2} \equiv \frac{\sqrt{6}-\sqrt{2}}{4}\end{aligned}$$

9 b iii)

$$\begin{aligned}\cos 105^\circ &\equiv \cos(60^\circ + 45^\circ)\\ &\equiv \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ\\ &\equiv \frac{1}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} \equiv \frac{\sqrt{2}-\sqrt{6}}{4}\end{aligned}$$

9 b iv)

$$\begin{aligned}\cos 15^\circ &\equiv \cos(60^\circ - 45^\circ)\\ &\equiv \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ\\ &\equiv \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} \equiv \frac{\sqrt{2}+\sqrt{6}}{4}\end{aligned}$$

Alternatively:

$$\begin{aligned}\cos 15^\circ &\equiv \cos(45^\circ - 30^\circ)\\ &\equiv \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ\\ &\equiv \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2} \equiv \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

10 a)

$$\begin{aligned}\tan 30^\circ &= \frac{\sqrt{3}}{3}\\ \tan 45^\circ &= 1\\ \tan 60^\circ &= \sqrt{3}\end{aligned}$$

10 b i)

$$\begin{aligned}\tan 75^\circ \equiv \tan(30^\circ + 45^\circ) &\equiv \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ} \equiv \frac{\frac{\sqrt{3}}{3} + 1}{1 - \frac{\sqrt{3}}{3} \times 1}\\ &\equiv \frac{\sqrt{3}+3}{3-\sqrt{3}} \equiv \frac{\sqrt{3}+3}{3-\sqrt{3}} \times \frac{3+\sqrt{3}}{3+\sqrt{3}}\\ &\equiv \frac{9 + 2 \times 3\sqrt{3} + 3}{9-3} \equiv \frac{12+6\sqrt{3}}{6} \equiv 2 + \sqrt{3}\end{aligned}$$

10 b ii)

$$\begin{aligned}\tan 15^\circ \equiv \tan(60^\circ - 45^\circ) &\equiv \frac{\tan 60^\circ - \tan 45^\circ}{1 + \tan 60^\circ \tan 45^\circ} \\ &\equiv \frac{\sqrt{3} - 1}{1 + \sqrt{3} \times 1} \equiv \frac{\sqrt{3} - 1}{1 + \sqrt{3}} \times \frac{1 - \sqrt{3}}{1 - \sqrt{3}} \\ &\equiv \frac{-1 + 2\sqrt{3} - 3}{1 - 3} \equiv \frac{2\sqrt{3} - 4}{-2} \equiv 2 - \sqrt{3}\end{aligned}$$

Alternatively:

$$\begin{aligned}\tan 15^\circ \equiv \tan(45^\circ - 30^\circ) &\equiv \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} \equiv \frac{1 - \frac{\sqrt{3}}{3}}{1 + 1 \times \frac{\sqrt{3}}{3}} \\ &\equiv \frac{3 - \sqrt{3}}{3 + \sqrt{3}} \equiv \frac{3 - \sqrt{3}}{3 + \sqrt{3}} \times \frac{3 - \sqrt{3}}{3 - \sqrt{3}} \\ &\equiv \frac{9 - 2 \times 3\sqrt{3} + 3}{9 - 3} \equiv \frac{12 - 6\sqrt{3}}{6} \equiv 2 - \sqrt{3}\end{aligned}$$

11 a)

$$\begin{aligned}\sin(x + 90^\circ) &\equiv \sin x \cos 90^\circ + \cos x \sin 90^\circ \\ &\equiv \sin x \times 0 + \cos x \times 1 \equiv \cos x\end{aligned}$$

11 b)

$$\begin{aligned}\cos(x - 90^\circ) &\equiv \cos x \cos 90^\circ + \sin x \sin 90^\circ \\ &\equiv \cos x \times 0 + \sin x \times 1 \equiv \sin x\end{aligned}$$

11 c)

$$\begin{aligned}\cos(x - 180^\circ) &\equiv \cos x \cos 180^\circ + \sin x \sin 180^\circ \\ &\equiv \cos x \times -1 + \sin x \times 0 \equiv -\cos x\end{aligned}$$

11 d)

$$\begin{aligned}\sin(x - 270^\circ) &\equiv \sin x \cos 270^\circ - \cos x \sin 270^\circ \\ &\equiv \sin x \times 0 - \cos x \times -1 \equiv \cos x\end{aligned}$$

12 a)

$$\begin{aligned}&\cos 40^\circ \cos 32^\circ - \sin 40^\circ \sin 32^\circ \\ &\qquad\qquad \equiv \cos(40^\circ + 32^\circ) \equiv \cos 72^\circ\end{aligned}$$

As we saw in11)a), $\cos x \equiv \sin(x + 90^\circ)$.

$$\therefore \cos 72^\circ \equiv \sin(72^\circ + 90^\circ) \equiv \sin 162^\circ \equiv \sin 18^\circ$$

12 b)

$$\cos 128^\circ \cos 125^\circ + \sin 128^\circ \sin 125^\circ$$
$$\equiv \cos(128^\circ - 125^\circ) \equiv \cos 3^\circ$$

As we saw in11)a), $\cos x \equiv \sin(x + 90^\circ)$.

$$\therefore \cos 3^\circ \equiv \sin(3^\circ + 90^\circ) \equiv \sin 93^\circ \equiv \sin 87^\circ$$

13 a)

$$\sin 96^\circ \cos 5^\circ - \cos 96^\circ \sin 5^\circ$$
$$\equiv \sin(96^\circ - 5^\circ) \equiv \sin 91^\circ$$

As we saw in11)b), $\sin x \equiv \cos(x - 90^\circ)$.

$$\therefore \sin 91^\circ \equiv \cos(91^\circ - 90^\circ) \equiv \cos 1^\circ$$

13 b)

$$\sin 25^\circ \cos 42^\circ + \cos 25^\circ \sin 42^\circ$$
$$\equiv \sin(25^\circ + 42^\circ) \equiv \sin 67^\circ$$

As we saw in11)b), $\sin x \equiv \cos(x - 90^\circ)$.

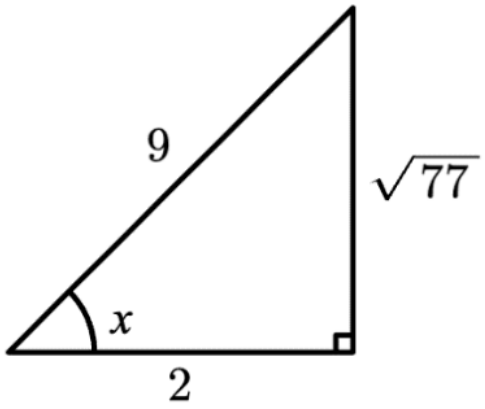
$$\therefore \sin 67^\circ \equiv \cos(67^\circ - 90^\circ) \equiv \cos(-23^\circ) \equiv \cos 23^\circ$$

14 a)

$\cos x = \frac{2}{9}$.

x is **ACUTE**.

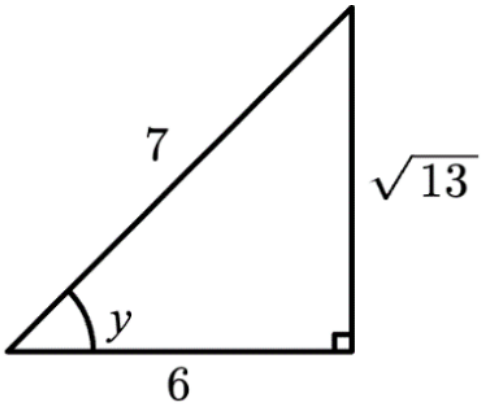
Draw a triangle with this information:



$$\sin x = \frac{\sqrt{77}}{9}$$
$$\cos y = \frac{6}{7}$$

y is **ACUTE**.

Draw a triangle with this information:



$$\sin y = \frac{\sqrt{13}}{7}$$
$$\therefore \cos(x-y) \equiv \cos x \cos y + \sin x \sin y$$
$$\equiv \frac{2}{9} \times \frac{6}{7} + \frac{\sqrt{77}}{9} \times \frac{\sqrt{13}}{7} \equiv \frac{12+\sqrt{1001}}{63}$$

14 b)

$$\sin(x-y) \equiv \sin x \cos y - \cos x \sin y$$
$$\equiv \frac{\sqrt{77}}{9} \times \frac{6}{7} - \frac{2}{9} \times \frac{\sqrt{13}}{7} \equiv \frac{6\sqrt{77}-2\sqrt{13}}{63}$$

15)

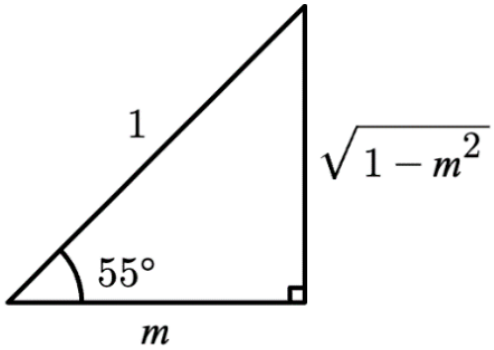
$$\begin{aligned}\tan 130^\circ &\equiv \tan(85^\circ + 45^\circ) \equiv \frac{\tan 85^\circ + \tan 45^\circ}{1 - \tan 85^\circ \tan 45^\circ} \\ &\equiv \frac{b+1}{1 - b \times 1} \equiv \frac{1+b}{1-b}\end{aligned}$$

16)

$$\begin{aligned}\cos 5^\circ &\equiv \cos(60^\circ - 55^\circ) \\ &\equiv \cos 60^\circ \cos 55^\circ + \sin 60^\circ \sin 55^\circ \\ \cos 55^\circ &= m = \frac{m}{1}\end{aligned}$$

55° is **ACUTE**.

Draw a triangle with this information:



$$\begin{aligned}\sin 55^\circ &= \frac{\sqrt{1-m^2}}{1} = \sqrt{1-m^2} \\ \therefore \cos 5^\circ &= \frac{1}{2} \times m + \frac{\sqrt{3}}{2} \times \sqrt{1-m^2} \\ &= \frac{m + \sqrt{3}\sqrt{1-m^2}}{2} = \frac{m + \sqrt{3-3m^2}}{2}\end{aligned}$$

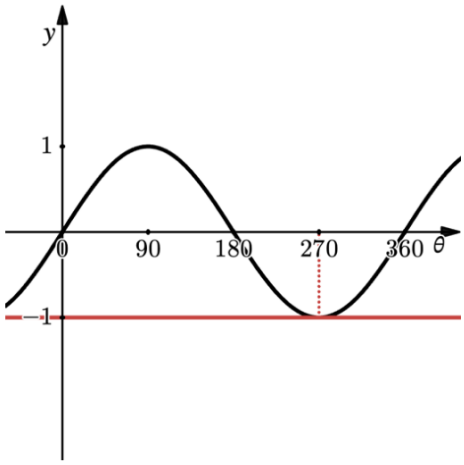
17 a)

$$\sin\theta\cos 60^\circ + \cos\theta\sin 60^\circ \equiv \sin(\theta + 60^\circ)$$

This has a minimum value of -1, as $y = \sin(\theta + 60^\circ)$ has a range of $-1 \leq y \leq 1$.

Solve $\sin(\theta + 60^\circ) = -1$:

$$\theta + 60^\circ = \arcsin(-1) = -90^\circ$$



$$\theta + 60^\circ = -90^\circ + 360^\circ = 270^\circ$$

Apply transformation:

$$\theta + 60^\circ = -90^\circ \Rightarrow \theta = -150^\circ$$
$$\theta + 60^\circ = 270^\circ \Rightarrow \theta = 210^\circ$$

So $\theta = 210^\circ$ is the first value of θ for which the minimum occurs.

$$\therefore \text{Minimum} = -1 \text{ when } \theta = 210^\circ$$

17 b)

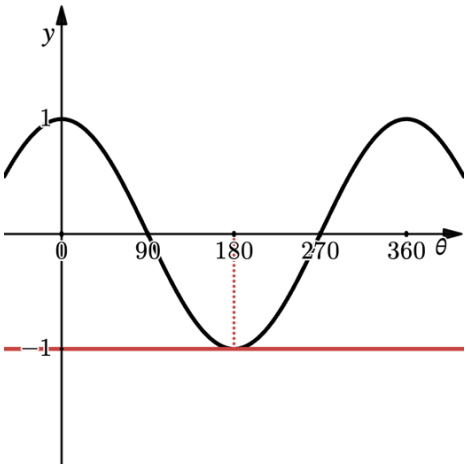
$2\cos\theta\cos 35° + 2\sin\theta\sin 35°$

$$\equiv 2(\cos\theta\cos 35° + \sin\theta\sin 35°) \equiv 2\cos(\theta - 35°)$$

This has a minimum value of -2, as $y = 2\cos(\theta - 35°)$ has a range of $-2 \leq y \leq 2$.

Solve $2\cos(\theta - 35°) = -2 \Rightarrow \cos(\theta - 35°) = -1$.

$$\theta - 35° = \arccos(-1) = 180°$$



Apply transformation: $\theta - 35° = 180° \Rightarrow \theta = 215°$.

So $\theta = 215°$ is the first value of θ for which the minimum occurs.

$\therefore$ Minimum $= -2$ when $\theta = 215°$.

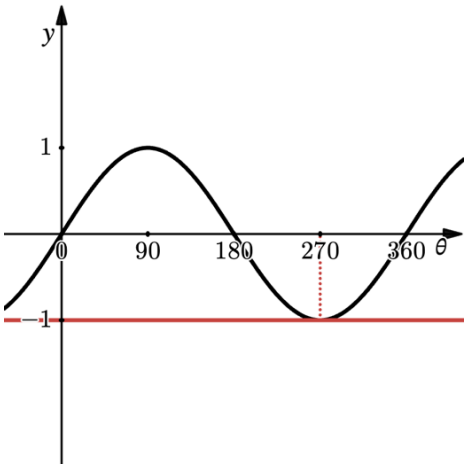
17 c)

$4\sin\theta\cos 42° - 4\cos\theta\sin 42° \equiv 4(\sin\theta\cos 42° - \cos\theta\sin 42°) \equiv 4\sin(\theta - 42°)$.

This has a minimum value of -4, as $y = 4\sin(\theta - 42°)$ has a range of $-4 \leq y \leq 4$.

Solve $4\sin(\theta - 42°) = -4 \Rightarrow \sin(\theta - 42°) = -1$.

$$\theta - 42° = \arcsin(-1) = -90°$$



$\theta - 42° = -90° + 360° = 270°$.

Apply transformation:

$$\theta - 42° = -90° \Rightarrow \theta = -48°$$

$$\theta - 42° = 270° \Rightarrow \theta = 312°$$

So $\theta = 312°$ is the first value of θ for which the minimum occurs.

$\therefore$ Minimum $= -4$ when $\theta = 312°$.

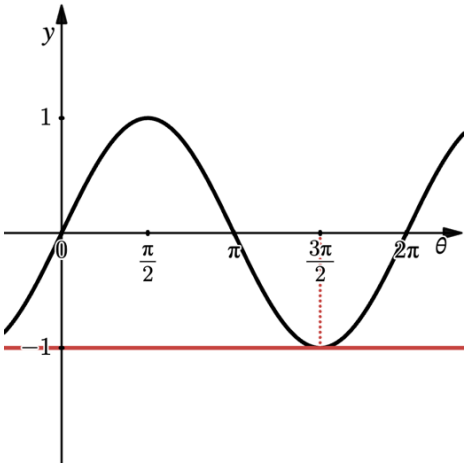
18 a)

$\sin\theta\cos\frac{\pi}{4} + \cos\theta\sin\frac{\pi}{4} \equiv \sin(\theta + \frac{\pi}{4})$.

This has a minimum value of -1, as $y = \sin(\theta + \frac{\pi}{4})$ has a range of $-1 \leq y \leq 1$.

Solve $\sin(\theta + \frac{\pi}{4}) = -1$.

$$\theta + \frac{\pi}{4} = \arcsin(-1) = -\frac{\pi}{2}$$



$\theta + \frac{\pi}{4} = -\frac{\pi}{2} + 2\pi = \frac{3\pi}{2}$.

Apply transformation:

$$\theta + \frac{\pi}{4} = -\frac{\pi}{2} \Rightarrow \theta = -\frac{\pi}{4}$$

$$\theta + \frac{\pi}{4} = \frac{3\pi}{2} \Rightarrow \theta = \frac{5\pi}{4}$$

So $\theta = \frac{5\pi}{4}$ is the first value of θ for which the minimum occurs.

$\therefore$ Minimum $= -1$ when $\theta = \frac{5\pi}{4}$.

18 b)

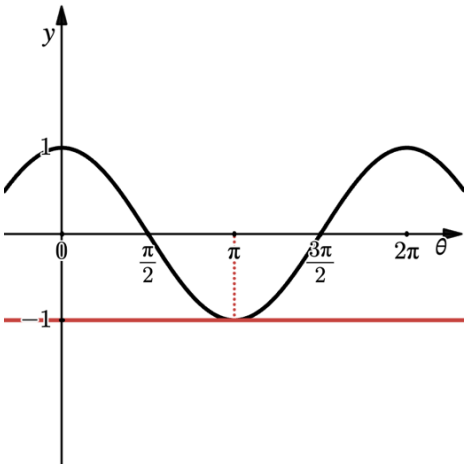
$3\cos\theta\cos\frac{5\pi}{4} + 3\sin\theta\sin\frac{5\pi}{4}$

$$\equiv 3(\cos\theta\cos\tfrac{5\pi}{4} + \sin\theta\sin\tfrac{5\pi}{4}) \equiv 3\cos(\theta - \tfrac{5\pi}{4})$$

This has a minimum value of -3, as $y = 3\cos(\theta - \frac{5\pi}{4})$ has a range of $-3 \leq y \leq 3$.

Solve $\cos(\theta - \frac{5\pi}{4}) = -1$.

$$\theta - \frac{5\pi}{4} = \arccos(-1) = \pi$$



Apply transformation: $\theta - \frac{5\pi}{4} = \pi \Rightarrow \theta = \frac{9\pi}{4}$.

Subtract period: $\theta = \frac{9\pi}{4} - 2\pi = \frac{\pi}{4}$.

So $\theta = \frac{\pi}{4}$ is the first value of θ for which the minimum occurs.

$\therefore$ Minimum $= -3$ when $\theta = \frac{\pi}{4}$.

18 c)

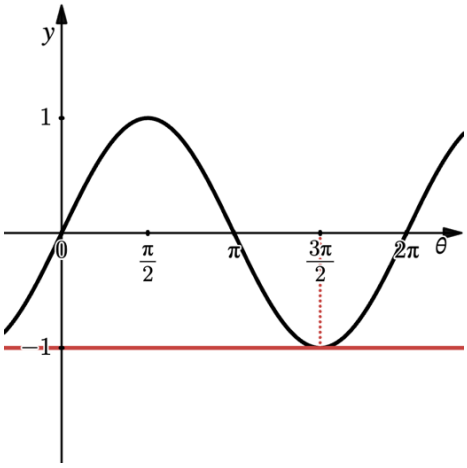
$-2\cos\theta\sin 1.6 + 2\sin\theta\cos 1.6$

$$\equiv 2(\sin\theta\cos 1.6 - \cos\theta\sin 1.6) \equiv 2\sin(\theta - 1.6)$$

This has a minimum value of -2, as $y = 2\sin(\theta - 1.6)$ has a range of $-2 \leq y \leq 2$.

Solve $2\sin(\theta - 1.6) = -2 \Rightarrow \sin(\theta - 1.6) = -1$.

$$\theta - 1.6 = \arcsin(-1) = -\frac{\pi}{2}$$



$\theta - 1.6 = -\frac{\pi}{2} + 2\pi = \frac{3\pi}{2}$.

Apply transformation:

$$\theta - 1.6 = -\frac{\pi}{2} \Rightarrow \theta = 1.6 - \frac{\pi}{2} = 0.0292\ldots$$

$$\theta - 1.6 = \frac{3\pi}{2} \Rightarrow \theta = 1.6 + \frac{3\pi}{2} = 6.3123\ldots$$

So $\theta = 1.6 - \frac{\pi}{2} = 0.0292$ to 3sf is the first value of θ for which the minimum occurs.

$\therefore$ Minimum $= -2$ when $\theta = 0.0292$.

19 a)

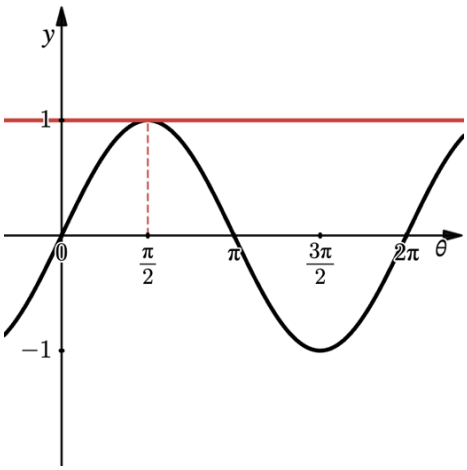
$\sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$

$$\equiv \sin(3\theta + 2\theta) \equiv \sin 5\theta$$

This has a maximum value of 1, as $y = \sin 5\theta$ has a range of $-1 \leq y \leq 1$.

Solve $\sin 5\theta = 1$.

$$5\theta = \arcsin 1 = \frac{\pi}{2}$$



Apply transformation: $5\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{10}$.

So $\theta = \frac{\pi}{10}$ is the first value of θ for which the maximum occurs.

$\therefore$ Maximum $= 1$ when $\theta = \frac{\pi}{10}$.

19 b)

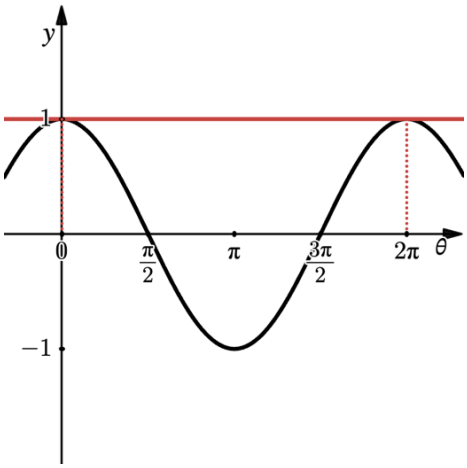
$9\cos 5\theta \cos 2\theta + 9\sin 5\theta \sin 2\theta$

$$\equiv 9(\cos 5\theta \cos 2\theta + \sin 5\theta \sin 2\theta) \equiv 9\cos(5\theta - 2\theta) \equiv 9\cos 3\theta$$

This has a maximum value of 9, as $y = 9\cos 3\theta$ has a range of $-9 \leq y \leq 9$.

Solve $9\cos 3\theta = 9 \Rightarrow \cos 3\theta = 1$.

$$3\theta = \arccos 1 = 0$$



Apply transformation: $3\theta = 0 \Rightarrow \theta = 0$.

But θ is positive, so we must add period:

$$\theta = 0 + \frac{2p}{3} = \frac{2\pi}{3}$$

So $\theta = \frac{2\pi}{3}$ is the first value of θ for which the maximum occurs.

$\therefore$ Maximum $= 9$ when $\theta = \frac{2\pi}{3}$.

19 c)

$-\frac{2}{3}\cos 8\theta \sin 2\theta + \frac{2}{3}\sin 8\theta \cos 2\theta$

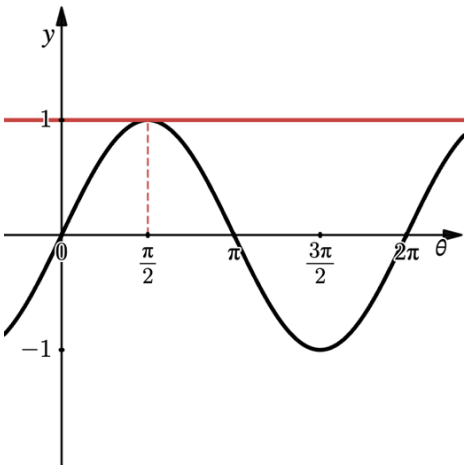
$$\equiv \frac{2}{3}(\sin 8\theta \cos 2\theta - \cos 8\theta \sin 2\theta)$$

$$\equiv \frac{2}{3}\sin(8\theta - 2\theta) = \frac{2}{3}\sin 6\theta$$

This has a maximum value of $\frac{2}{3}$, as $y = \frac{2}{3}\sin 6\theta$ has a range of $-\frac{2}{3} \leq y \leq \frac{2}{3}$.

Solve $\frac{2}{3}\sin 6\theta = \frac{2}{3} \Rightarrow \sin 6\theta = 1$.

$$6\theta = \arcsin 1 = \frac{\pi}{2}$$



Apply transformation: $6\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{12}$.

So $\theta = \frac{\pi}{12}$ is the first value of θ for which the maximum occurs.

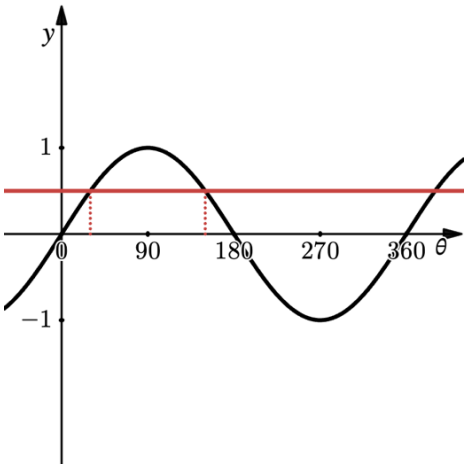
$\therefore$ Maximum $= \frac{2}{3}$ when $\theta = \frac{\pi}{12}$.

20 a)

$\sin\theta\cos 20° + \cos\theta\sin 20° = 0.5$ for $0° \leq \theta < 360°$.

$$\Rightarrow \sin(\theta + 20°) = 0.5$$

Find first value between $0°$ and $360°$: $\theta + 20° = \arcsin 0.5 = 30°$.



Find second value between $0°$ and $360°$: $\theta + 20° = 180° - 30° = 150°$.

Apply transformation:

$$\theta + 20° = 30° \Rightarrow \theta = 10° \quad \text{(in range)}$$

$$\theta + 20° = 150° \Rightarrow \theta = 130° \quad \text{(in range)}$$

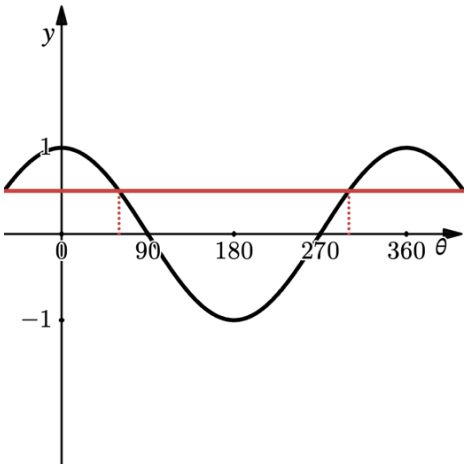
$\theta = \{10°, 130°\}$.

20 b)

$\cos\theta\cos 40° + \sin\theta\sin 40° = 0.34$ for $0° \leq \theta < 360°$.

$$\Rightarrow \cos(\theta - 40°) = 0.34$$

Find first value between $0°$ and $360°$: $\theta - 40° = \arccos 0.34 = 70.123\ldots°$.



Find second value between $0°$ and $360°$: $\theta - 40° = 360° - 70.123\ldots° = 289.876\ldots°$.

Apply transformation:

$$\theta - 40° = 70.123\ldots° \Rightarrow \theta = 110.123\ldots° \quad \text{(in range)}$$

$$\theta - 40° = 289.876\ldots° \Rightarrow \theta = 329.876\ldots° \quad \text{(in range)}$$

$\theta = \{110°, 330°\}$ to 3sf.

21 a)

$$\frac{\tan 200^\circ + \tan 34^\circ}{1 - \tan 200^\circ \tan 34^\circ} \equiv \tan(200^\circ + 34^\circ)$$
$$= \tan 234^\circ = \tan 54^\circ$$

21 b)

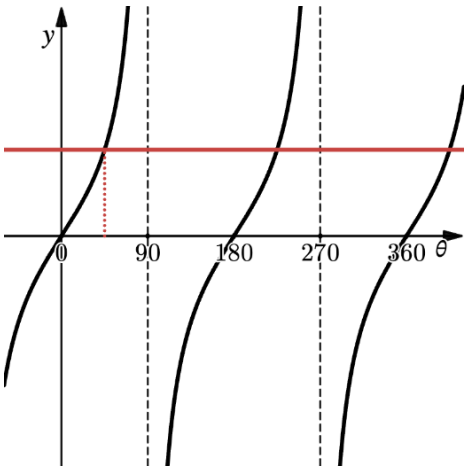
$$\frac{\tan 18^\circ - \tan 130^\circ}{1 + \tan 18^\circ \tan 130^\circ} \equiv \tan(18^\circ - 130^\circ)$$
$$= \tan(-112^\circ) = \tan 68^\circ$$

22 a)

$\frac{\tan\theta + \tan 20^\circ}{1 - \tan\theta \tan 20^\circ} = 4$ for $0^\circ \leq \theta < 360^\circ$.

$$\Rightarrow \tan(\theta + 20^\circ) = 4$$

Find first value: $\theta + 20^\circ = \arctan 4 = 75.963\ldots^\circ$.



Apply transformation: $\theta + 20^\circ = 75.963\ldots^\circ \Rightarrow \theta = 55.963\ldots^\circ$ (in range).

Add period: $\theta = 55.963\ldots^\circ + 180^\circ = 235.963\ldots^\circ$ (in range).

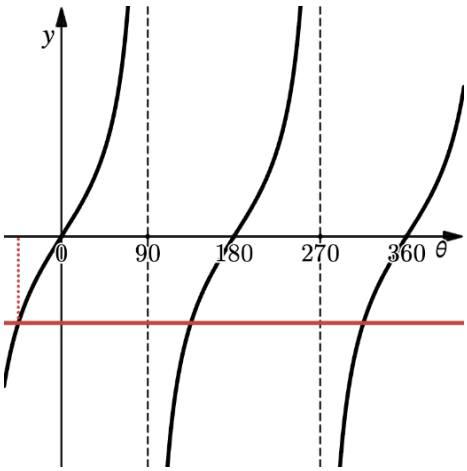
$\theta = \{56.0^\circ, 236^\circ\}$ to 3sf.

22 b)

$\frac{\tan 3\theta - \tan 33^\circ}{1+\tan 3\theta \tan 33^\circ} = -6$ for $0^\circ \leq \theta < 120^\circ$.

$$\Rightarrow \tan(3\theta - 33^\circ) = -6$$

Find first value: $3\theta - 33^\circ = \arctan(-6) = -80.537\ldots^\circ$.



Apply transformation: $3\theta - 33^\circ = -80.537\ldots^\circ \Rightarrow \theta = -15.845\ldots^\circ$ (not in range).

Add period:

$$\theta = -15.845\ldots^\circ + 60^\circ = 44.154\ldots^\circ \quad \text{(in range)}$$

$$\theta = 44.154\ldots^\circ + 60^\circ = 104.154\ldots^\circ \quad \text{(in range)}$$

$\theta = \{44.2^\circ, 104^\circ\}$ to 3sf.

23 a)

$\sin(x + 45°) + \cos(x + 45°)$

$$\equiv (\sin x \cos 45° + \cos x \sin 45°) + (\cos x \cos 45° - \sin x \sin 45°)$$

$$\equiv \frac{\sqrt{2}}{2}\sin x + \frac{\sqrt{2}}{2}\cos x + \frac{\sqrt{2}}{2}\cos x - \frac{\sqrt{2}}{2}\sin x \equiv \sqrt{2}\cos x$$

23 b)

$\sin(x + 135°) - \cos(x + 135°)$

$$\equiv (\sin x \cos 135° + \cos x \sin 135°) - (\cos x \cos 135° - \sin x \sin 135°)$$

$$\equiv -\frac{\sqrt{2}}{2}\sin x + \frac{\sqrt{2}}{2}\cos x + \frac{\sqrt{2}}{2}\cos x + \frac{\sqrt{2}}{2}\sin x$$

$$\equiv \sqrt{2}\cos x$$

23 c)

$\sin(x - 75°) - \cos(x - 75°)$

$$\equiv (\sin x \cos 75° - \cos x \sin 75°) - (\cos x \cos 75° + \sin x \sin 75°)$$

$$\equiv \sin x \times \frac{\sqrt{6} - \sqrt{2}}{4} - \cos x \times \frac{\sqrt{6} + \sqrt{2}}{4} - \cos x \times \frac{\sqrt{6} - \sqrt{2}}{4} - \sin x \times \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\equiv -\frac{\sqrt{2}}{2}\sin x - \frac{\sqrt{6}}{2}\cos x$$

24 a)

$\sin(x - \frac{\pi}{3}) + \cos(\frac{\pi}{3} - x)$

$$\equiv (\sin x \cos \tfrac{\pi}{3} - \cos x \sin \tfrac{\pi}{3}) + (\cos \tfrac{\pi}{3} \cos x + \sin \tfrac{\pi}{3} \sin x)$$

$$\equiv \frac{1}{2} \sin x - \frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x$$

$$\equiv \frac{1 + \sqrt{3}}{2} \sin x + \frac{1 - \sqrt{3}}{2} \cos x$$

24 b)

$\sin(\frac{\pi}{3} - x) - \cos(\frac{4\pi}{3} - x)$

$$\equiv (\sin \tfrac{\pi}{3} \cos x - \cos \tfrac{\pi}{3} \sin x) - (\cos \tfrac{4\pi}{3} \cos x + \sin \tfrac{4\pi}{3} \sin x)$$

$$\equiv \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x + \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x$$

$$\equiv \frac{\sqrt{3} + 1}{2} \cos x + \frac{\sqrt{3} - 1}{2} \sin x$$

24 c)

$-\sin(\frac{\pi}{4} - x) + \cos(\frac{2\pi}{3} - x)$

$$\equiv -(\sin \tfrac{\pi}{4} \cos x - \cos \tfrac{\pi}{4} \sin x) + (\cos \tfrac{2\pi}{3} \cos x + \sin \tfrac{2\pi}{3} \sin x)$$

$$\equiv -(\tfrac{\sqrt{2}}{2} \cos x - \tfrac{\sqrt{2}}{2} \sin x) + (-\tfrac{1}{2} \cos x + \tfrac{\sqrt{3}}{2} \sin x)$$

$$\equiv -\frac{\sqrt{2}}{2} \cos x + \frac{\sqrt{2}}{2} \sin x - \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x$$

$$\equiv \frac{\sqrt{2} + \sqrt{3}}{2} \sin x - \frac{1 + \sqrt{2}}{2} \cos x$$

25 a)

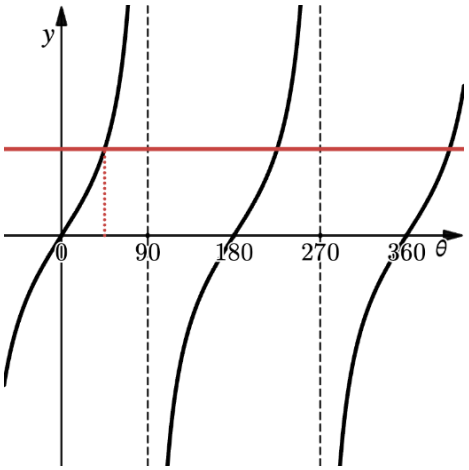
$\sin(\theta + 30°) = \cos\theta$ for $0° < \theta < 360°$.

$$\Rightarrow \sin\theta\cos 30° + \cos\theta\sin 30° = \cos\theta$$

$$\Rightarrow \frac{\sqrt{3}}{2}\sin\theta + \frac{1}{2}\cos\theta = \cos\theta \Rightarrow \frac{\sqrt{3}}{2}\sin\theta = \frac{1}{2}\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{3}} \Rightarrow \tan\theta = \frac{\sqrt{3}}{3}$$

Find first value: $\theta = \arctan\frac{\sqrt{3}}{3} = 30°$ (in range).



Add period: $\theta = 30° + 180° = 210°$ (in range).

$\Rightarrow \theta = \{30°, 210°\}$.

25 b)

$\sin(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$.

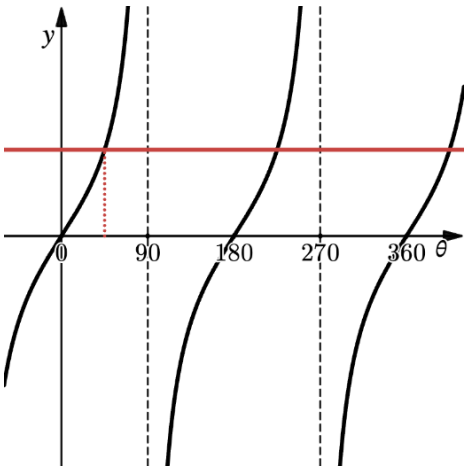
$$\Rightarrow \sin\theta\cos 30° + \cos\theta\sin 30° = \sin\theta$$

$$\Rightarrow \frac{\sqrt{3}}{2}\sin\theta + \frac{1}{2}\cos\theta = \sin\theta$$

$$\Rightarrow \frac{1}{2}\cos\theta = (1 - \tfrac{\sqrt{3}}{2})\sin\theta \Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\frac{1}{2}}{1 - \frac{\sqrt{3}}{2}}$$

$$\Rightarrow \tan\theta = 2 + \sqrt{3}$$

Find first value: $\theta = \arctan(2 + \sqrt{3}) = 75°$ (in range).



Add period: $\theta = 75° + 180° = 255°$ (in range).

$\Rightarrow \theta = \{75°, 255°\}$.

25 c)

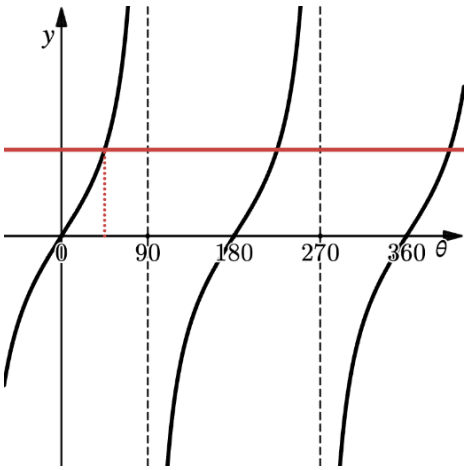
$\cos(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$.

$$\Rightarrow \cos\theta\cos 30° - \sin\theta\sin 30° = \sin\theta$$

$$\Rightarrow \frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta = \sin\theta \Rightarrow \frac{\sqrt{3}}{2}\cos\theta = \frac{3}{2}\sin\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{3}}{3} \Rightarrow \tan\theta = \frac{\sqrt{3}}{3}$$

Find first value: $\theta = \arctan\frac{\sqrt{3}}{3} = 30°$ (in range).



Add period: $\theta = 30° + 180° = 210°$ (in range).

$\Rightarrow \theta = \{30°, 210°\}$.

26 a)

$\sin(\theta - 30°) = \cos(\theta + 45°)$ for $0° < \theta < 360°$.

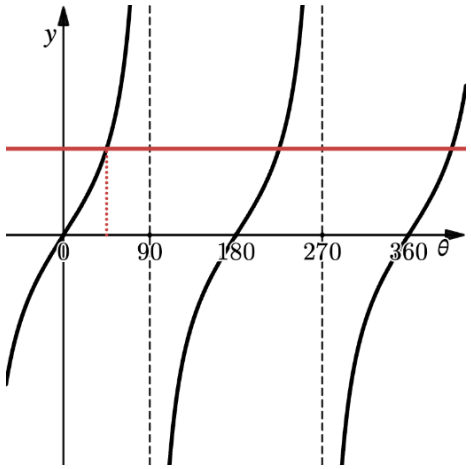
$$\Rightarrow \sin\theta\cos 30° - \cos\theta\sin 30° = \cos\theta\cos 45° - \sin\theta\sin 45°$$

$$\Rightarrow \frac{\sqrt{3}}{2}\sin\theta - \frac{1}{2}\cos\theta = \frac{\sqrt{2}}{2}\cos\theta - \frac{\sqrt{2}}{2}\sin\theta$$

$$\Rightarrow (\tfrac{\sqrt{3}}{2} + \tfrac{\sqrt{2}}{2})\sin\theta = (\tfrac{\sqrt{2}}{2} + \tfrac{1}{2})\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\frac{\sqrt{2}}{2} + \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}} \Rightarrow \tan\theta = \frac{\sqrt{2}+1}{\sqrt{3}+\sqrt{2}}$$

Find first value: $\theta = \arctan(\frac{\sqrt{2}+1}{\sqrt{3}+\sqrt{2}}) = 37.5°$ (in range).



Add period: $\theta = 37.5° + 180° = 217.5°$ (in range).

$\Rightarrow \theta = \{37.5°, 217.5°\}$.

26 b)

$\cos(\theta - 60°) = \cos(\theta - 45°)$ for $0° < \theta < 360°$.

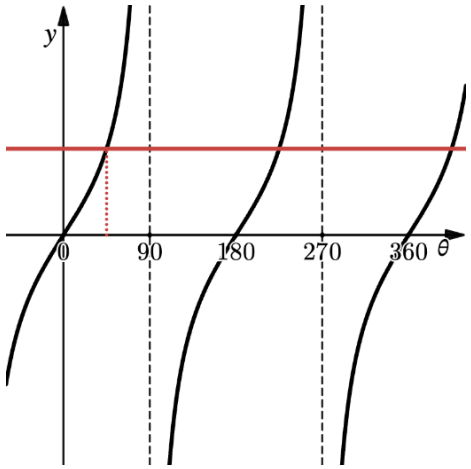
$$\Rightarrow \cos\theta\cos 60° + \sin\theta\sin 60° = \cos\theta\cos 45° + \sin\theta\sin 45°$$

$$\Rightarrow \frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta = \frac{\sqrt{2}}{2}\cos\theta + \frac{\sqrt{2}}{2}\sin\theta$$

$$\Rightarrow (\tfrac{\sqrt{3}}{2} - \tfrac{\sqrt{2}}{2})\sin\theta = (\tfrac{\sqrt{2}}{2} - \tfrac{1}{2})\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\frac{\sqrt{2}}{2} - \frac{1}{2}}{\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}} \Rightarrow \tan\theta = \frac{\sqrt{2} - 1}{\sqrt{3} - \sqrt{2}}$$

Find first value: $\theta = \arctan(\frac{\sqrt{2}-1}{\sqrt{3}-\sqrt{2}}) = 52.5°$ (in range).



Add period: $\theta = 52.5° + 180° = 232.5°$ (in range).

$\Rightarrow \theta = \{52.5°, 233°\}$ to 3sf.

26 c)

$\cos(\theta - 120°) = \sin(\theta + 60°)$ for $0° < \theta < 360°$.

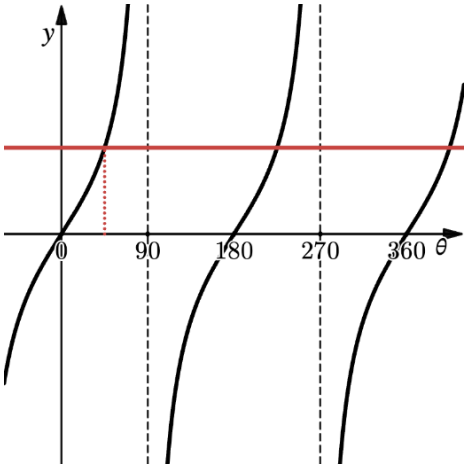
$$\Rightarrow \cos\theta\cos 120° + \sin\theta\sin 120° = \sin\theta\cos 60° + \cos\theta\sin 60°$$

$$\Rightarrow -\frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta = \frac{1}{2}\sin\theta + \frac{\sqrt{3}}{2}\cos\theta$$

$$\Rightarrow (\tfrac{\sqrt{3}}{2} - \tfrac{1}{2})\sin\theta = (\tfrac{\sqrt{3}}{2} + \tfrac{1}{2})\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\frac{\sqrt{3}}{2} + \frac{1}{2}}{\frac{\sqrt{3}}{2} - \frac{1}{2}} \Rightarrow \tan\theta = \frac{\sqrt{3}+1}{\sqrt{3}-1} \Rightarrow \tan\theta = 2 + \sqrt{3}$$

Find first value: $\theta = \arctan(2 + \sqrt{3}) = 75°$ (in range).



Add period: $\theta = 75° + 180° = 255°$ (in range).

$\Rightarrow \theta = \{75°, 255°\}$.

27)

$\sin x = \frac{\sqrt{2}}{BC}$

$\cos x = \frac{AB}{BC}$

$\sin y = \frac{\sqrt{5}}{BD}$

$\cos y = \frac{BC}{BD}$

$$\sin(x+y) = \frac{DE}{BD} \Rightarrow \sin x \cos y + \cos x \sin y = \frac{DE}{BD}$$

$$\Rightarrow \frac{\sqrt{2}}{BC} \times \frac{BC}{BD} + \cos x \times \frac{\sqrt{5}}{BD} = \frac{DE}{BD}$$

$$\Rightarrow \frac{\sqrt{2}}{BD} + \cos x \times \frac{\sqrt{5}}{BD} = \frac{DE}{BD} \Rightarrow \sqrt{2} + \cos x \times \sqrt{5} = DE$$

$$\Rightarrow DE = \sqrt{2} + \sqrt{5}\cos x$$

28)

LHS: $\frac{\cos(x+y)}{\cos x \cos y} \equiv \frac{\cos x \cos y - \sin x \sin y}{\cos x \cos y}$

$$\equiv \frac{\cos x \cos y}{\cos x \cos y} - \frac{\sin x \sin y}{\cos x \cos y} \equiv 1 - \frac{\sin x \sin y}{\cos x \cos y}$$

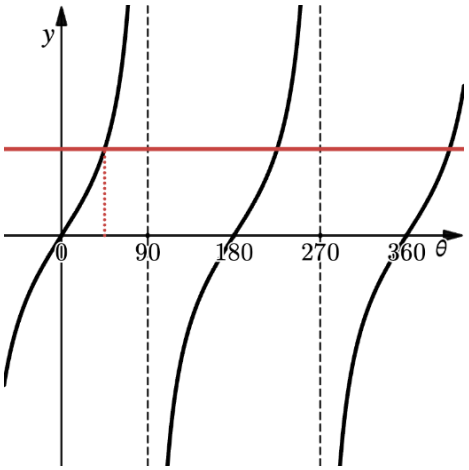
$$\equiv 1 - \frac{\sin x}{\cos x} \times \frac{\sin y}{\cos y} \equiv 1 - \tan x \tan y : \text{RHS}$$

29)

$\frac{\tan\theta - \tan 30°}{1+\tan\theta\tan 30°} = 2$ for $0° < \theta < 360°$.

$$\Rightarrow \tan(\theta - 30°) = 2$$

Find first value: $\theta - 30° = \arctan 2 = 63.434\ldots°$.



Apply transformation: $\theta - 30° = 63.434\ldots° \Rightarrow \theta = 93.434\ldots°$ (in range).

Add period: $\theta = 93.434\ldots° + 180° = 273.434\ldots°$ (in range).

$\theta = \{93.4°, 273°\}$ to 3sf.

30 a)

$3\cos(x - 45°) = 2\sin(x + 60°)$ for $0° < x < 360°$.

$$\Rightarrow 3(\cos x\cos 45° + \sin x\sin 45°) = 2(\sin x\cos 60° + \cos x\sin 60°)$$

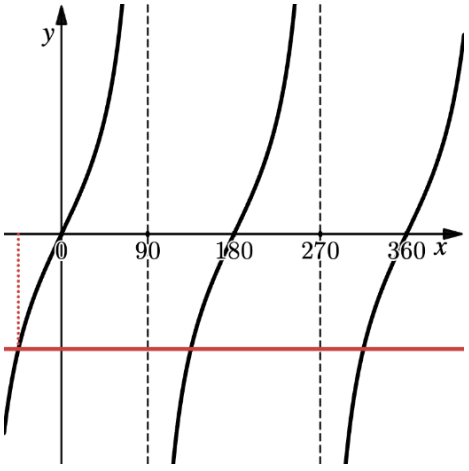
$$\Rightarrow 3\cos x\cos 45° + 3\sin x\sin 45° = 2\sin x\cos 60° + 2\cos x\sin 60°$$

$$\Rightarrow \frac{3\sqrt{2}}{2}\cos x + \frac{3\sqrt{2}}{2}\sin x = \sin x + \sqrt{3}\cos x$$

$$\Rightarrow (\tfrac{3\sqrt{2}}{2} - 1)\sin x = (\sqrt{3} - \tfrac{3\sqrt{2}}{2})\cos x$$

$$\Rightarrow \frac{\sin x}{\cos x} = \frac{\sqrt{3} - \frac{3\sqrt{2}}{2}}{\frac{3\sqrt{2}}{2} - 1} \Rightarrow \tan x = \frac{2\sqrt{3} - 3\sqrt{2}}{3\sqrt{2} - 2}$$

Find first value: $x = \arctan(\frac{2\sqrt{3}-3\sqrt{2}}{3\sqrt{2}-2}) = -19.144\ldots°$ (not in range).



Add period:

$$\theta = -19.144\ldots^\circ + 180^\circ = 160.855\ldots^\circ \quad \text{(in range)}$$

$$\theta = 160.855\ldots^\circ + 180^\circ = 340.855\ldots^\circ \quad \text{(in range)}$$

$\Rightarrow \theta = \{161^\circ, 341^\circ\}$ to 3sf.

30 b)

$3\cos(3\theta - 45^\circ) = 2\sin(3\theta + 60^\circ)$ for $0^\circ < \theta < 120^\circ$.

Apply transformation:

$$3\theta = 160.855\ldots^\circ \Rightarrow \theta = 53.618\ldots^\circ$$

$$3\theta = 340.855\ldots^\circ \Rightarrow \theta = 113.618\ldots^\circ$$

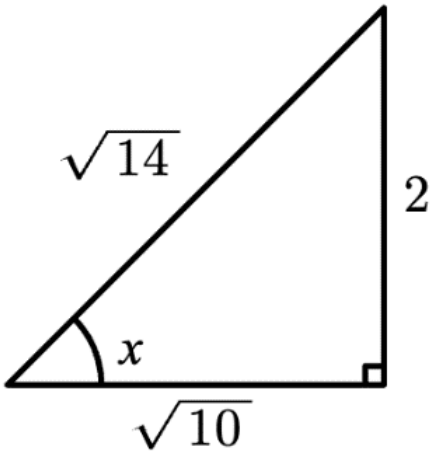
$\Rightarrow \theta = \{53.6^\circ, 114^\circ\}$ to 3sf.

31)

$\sin x = \frac{2}{\sqrt{14}}$.

x is **OBTUSE**.

Ignore $90^\circ < x < 180^\circ$ for the moment and draw a right-angled triangle with this information:

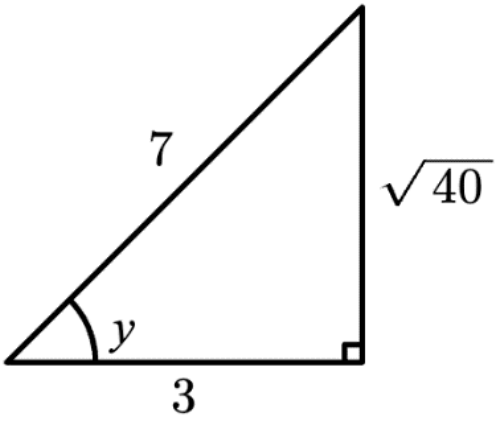


$\tan x = -\frac{2}{\sqrt{10}} = -\frac{\sqrt{10}}{5}$ (the result is negative as $\tan x$ is below the x-axis between 90° and 180°).

$\cos y = \frac{3}{7}$.

y is **ACUTE**.

Draw a triangle with this information:



$\tan y = \frac{\sqrt{40}}{3} = \frac{2\sqrt{10}}{3}$.

$$\tan(x-y) \equiv \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\Rightarrow \tan(x-y) = \frac{-\frac{\sqrt{10}}{5} - \frac{2\sqrt{10}}{3}}{1 + (-\frac{\sqrt{10}}{5})(\frac{2\sqrt{10}}{3})} = \frac{-\frac{13\sqrt{10}}{15}}{1 - \frac{4}{3}} = \frac{-\frac{13\sqrt{10}}{15}}{-\frac{1}{3}}$$

$$= \frac{13\sqrt{10}}{15} \times 3 = \frac{13\sqrt{10}}{5}$$

32)

$\tan(x+y) = 13 \Rightarrow \frac{\tan x + \tan y}{1 - \tan x \tan y} = 13$

$$\Rightarrow \frac{\frac{1}{2} + \tan y}{1 - \frac{1}{2}\tan y} = 13 \Rightarrow \frac{1 + 2\tan y}{2 - \tan y} = 13$$

$$\Rightarrow 1 + 2\tan y = 13(2 - \tan y)$$

$$\Rightarrow 1 + 2\tan y = 26 - 13\tan y \Rightarrow 15\tan y = 25$$

$$\Rightarrow \tan y = \frac{5}{3}$$

33)

LHS: $2\sin(x+45^\circ)\sin(x-45^\circ)$

$$\equiv 2(\sin x \cos 45^\circ + \cos x \sin 45^\circ)(\sin x \cos 45^\circ - \cos x \sin 45^\circ)$$

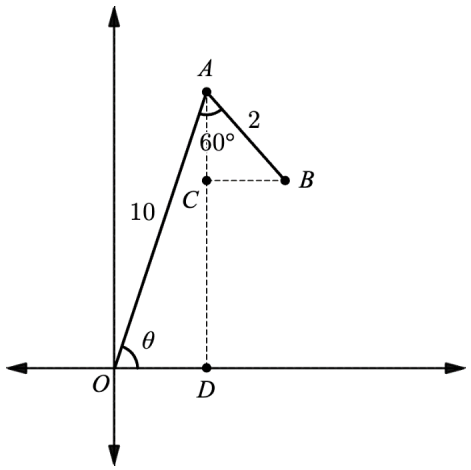
$$\equiv 2(\tfrac{\sqrt{2}}{2}\sin x + \tfrac{\sqrt{2}}{2}\cos x)(\tfrac{\sqrt{2}}{2}\sin x - \tfrac{\sqrt{2}}{2}\cos x)$$

$$\equiv 2 \times \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2}(\sin x + \cos x)(\sin x - \cos x)$$

$$\equiv (\sin x + \cos x)(\sin x - \cos x)$$

$$\equiv \sin^2 x - \cos^2 x : \text{RHS}$$

34)



$\sin\theta = \frac{AD}{10} \Rightarrow AD = 10\sin\theta.$

$\cos\theta = \frac{OD}{10} \Rightarrow OD = 10\cos\theta.$

$\angle OAD = 90° - \theta.$

$\angle OAB = 60° \Rightarrow \angle OAD + \angle CAB = 60°.$

$$\Rightarrow 90° - \theta + \angle CAB = 60° \Rightarrow \angle CAB = \theta - 30°$$

$\sin(\theta - 30°) = \frac{BC}{2} \Rightarrow BC = 2\sin(\theta - 30°).$

$\cos(\theta - 30°) = \frac{AC}{2} \Rightarrow AC = 2\cos(\theta - 30°).$

$$\therefore OD + BC = 10\cos\theta + 2\sin(\theta - 30°)$$

$$= 10\cos\theta + 2(\sin\theta\cos 30° - \cos\theta\sin 30°)$$

$$= 10\cos\theta + 2\sin\theta\cos 30° - 2\cos\theta\sin 30°$$

$$= 10\cos\theta + \sqrt{3}\sin\theta - \cos\theta = 9\cos\theta + \sqrt{3}\sin\theta$$

And $AD - AC = 10\sin\theta - 2\cos(\theta - 30°)$

$$= 10\sin\theta - 2(\cos\theta\cos 30° + \sin\theta\sin 30°)$$

$$= 10\sin\theta - 2\cos\theta\cos 30° - 2\sin\theta\sin 30°$$

$$= 10\sin\theta - \sqrt{3}\cos\theta - \sin\theta = 9\sin\theta - \sqrt{3}\cos\theta$$

$\therefore B$ has coordinates: $(9\cos\theta + \sqrt{3}\sin\theta, 9\sin\theta - \sqrt{3}\cos\theta).$

35)

$\cos(X+Y) = \cos X \cos Y - \sin X \sin Y$

$\cos(X-Y) = \cos X \cos Y + \sin X \sin Y$

$$\Rightarrow \cos(X+Y) - \cos(X-Y) = -2\sin X \sin Y$$

$A = X + Y$ and $B = X - Y$ so

$$\frac{A+B}{2} = \frac{2X}{2} = X \text{ and } \frac{A-B}{2} = \frac{2Y}{2} = Y$$

$\therefore \cos(A) - \cos(B) = -2\sin(\frac{A+B}{2})\sin(\frac{A-B}{2})$ as req.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h} = \lim_{h\to 0} \frac{-2\sin(\frac{x+h+x}{2})\sin(\frac{x+h-x}{2})}{h}$$

$$= \lim_{h\to 0} \frac{-2\sin(\frac{2x+h}{2})\sin(\frac{h}{2})}{h}$$

For small angles this gives

$$= \lim_{h\to 0} \frac{-2\sin(\frac{2x+h}{2}) \times \frac{h}{2}}{h} = \lim_{h\to 0} -2\sin(\frac{2x+h}{2}) \times (\frac{1}{2})$$

$$= \lim_{h\to 0} -\sin(\frac{2x+h}{2})$$

$$= -\sin(\frac{2x}{2}) = -\sin(x) \quad \text{as req.}$$

36)

$\sin(X+Y) = \sin X \cos Y + \cos X \sin Y$

$\sin(X-Y) = \sin X \cos Y - \cos X \sin Y$

So $\sin(X+Y) - \sin(X-Y) = 2\cos X \sin Y$.

$A = X + Y$ and $B = X - Y$ so

$$\frac{A+B}{2} = \frac{2X}{2} = X \text{ and } \frac{A-B}{2} = \frac{2Y}{2} = Y$$

$\therefore \sin(A) - \sin(B) = 2\cos(\frac{A+B}{2})\sin(\frac{A-B}{2})$ as req.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h\to 0} \frac{2\cos(\frac{x+h+x}{2})\sin(\frac{x+h-x}{2})}{h}$$

$$= \lim_{h\to 0} \frac{2\cos(\frac{2x+h}{2})\sin(\frac{h}{2})}{h}$$

For small angles this gives

$$= \lim_{h\to 0} \frac{2\cos(\frac{2x+h}{2}) \times \frac{h}{2}}{h} = \lim_{h\to 0} 2\cos(\frac{2x+h}{2}) \times (\tfrac{1}{2})$$

$$= \lim_{h\to 0} \cos(\tfrac{2x+h}{2})$$

$$= \cos(\tfrac{2x}{2}) = \cos(x) \quad \text{as req.}$$